

1. Determine the limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt[5]{x^2+|y|+1} - 1}{x^2+|y|} =$$

$$\frac{1}{5}$$

等价无穷小

$$\star (1+x)^u - 1 \sim ux$$

$$(x^2+|y|+1)^{\frac{1}{5}} - 1 \sim \frac{1}{5}(x^2+|y|)$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\frac{1}{5}(x^2+|y|)}{x^2+|y|} = \frac{1}{5}$$

等价无穷小

$$\star \begin{cases} 1 - \cos x \sim \frac{1}{2}x^2 \\ e^x - 1 \sim x \\ \ln(1+x) \sim x \\ (1+x)^u - 1 \sim ux \end{cases}$$

2. Given the function $w = \ln(\sqrt{x^2+1} + \sqrt{y^2+1} + \sqrt{z^2+1})$, compute

$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} + z \frac{\partial w}{\partial z} =$$

这=偏导数 是字母
不同形式问

$$\frac{\partial w}{\partial x} = \frac{\frac{1}{\sqrt{x^2+1}} \times x}{\sqrt{x^2+1} + \sqrt{y^2+1} + \sqrt{z^2+1}}$$

$$\frac{\partial w}{\partial y} = \frac{\frac{y}{\sqrt{y^2+1}}}{\sqrt{x^2+1} + \sqrt{y^2+1} + \sqrt{z^2+1}}$$

$$\frac{\partial w}{\partial z} = \frac{\frac{z}{\sqrt{z^2+1}}}{\sqrt{x^2+1} + \sqrt{y^2+1} + \sqrt{z^2+1}}$$

$$\text{TRA} = \frac{\frac{x^2}{\sqrt{x^2+1}} + \frac{y^2}{\sqrt{y^2+1}} + \frac{z^2}{\sqrt{z^2+1}}}{\sqrt{x^2+1} + \sqrt{y^2+1} + \sqrt{z^2+1}}$$

3. Suppose the equation $F\left(xy, \frac{z}{x^2+y^2}\right) = 0$ determines an implicit function $z = z(x, y)$ where $F(u, v)$ is differentiable. Then $dz =$ _____ $\rightarrow$ 隐函数求导

$$f(x, y) = F\left(xy, \frac{z}{x^2+y^2}\right) = 0$$

$$dz = \frac{dz}{dx} dx + \frac{dz}{dy} dy \quad z = z(x, y)$$

$$F(u, v) = F\left(xy, \frac{z}{x^2+y^2}\right)$$

$$\therefore u = xy$$

$$v = \frac{z}{x^2+y^2}$$

$$\frac{\partial u}{\partial x} = y \quad \frac{\partial u}{\partial y} = x$$

$$\frac{\partial v}{\partial x} = \frac{1}{x^2+y^2} \cdot \frac{\partial z}{\partial x} - \frac{2x \cdot z}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{1}{x^2+y^2} \cdot \frac{\partial z}{\partial y} - \frac{2y \cdot z}{(x^2+y^2)^2}$$

$$F_u y + F_v \left[\frac{1}{x^2+y^2} \frac{\partial z}{\partial x} - \frac{2xz}{(x^2+y^2)^2} \right]$$

$$\frac{\partial z}{\partial x} = \frac{2xz}{x^2+y^2} - \frac{F_u}{F_v} y(x^2+y^2)$$

$$\frac{\partial z}{\partial y} = \frac{2yz}{x^2+y^2} - \frac{F_u}{F_v} x(x^2+y^2)$$

$$dz = z_x dx + z_y dy$$

4. Given a function $z = f(xy^2, 3x - 2y)$ where $f(u, v)$ is of second order continuous partial derivatives,

then $\frac{\partial z}{\partial x} =$ _____.

普通求导链式法则

$$z = f(xy^2, 3x - 2y) = f(u, v)$$

$$u = xy^2$$

$$\frac{\partial u}{\partial x} = y^2$$

$$\frac{\partial u}{\partial y} = 2xy$$

$$v = 3x - 2y$$

$$\frac{\partial v}{\partial x} = 3$$

$$\frac{\partial v}{\partial y} = -2$$

$$\frac{dz}{dx} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= y^2 \frac{\partial f}{\partial u} + 3 \frac{\partial f}{\partial v}$$

5. Let C be the curve determined by the parametric equations $x = \cos(\pi t)$, $y = -2\sin(\pi t)$, $z = 3\arctan(2t)$. The equation of the tangent line to the curve C at the point when $t = \frac{1}{2}$ is _____. The equation of the normal plane is _____.

切线3条 法平面3条.

$$x' = -\pi \sin(\pi t)$$

$$y' = -2\pi \cos(\pi t)$$

$$z' = \frac{3}{1+(2t)^2} \times 2 = \frac{6}{1+4t^2}$$

$$\text{Let } t = \frac{1}{2} \quad x' = -\pi \quad y' = 0 \quad z' = 3$$

$$\text{切线向量 } \vec{T} = (x', y', z') = (-\pi, 0, 3)$$

$$t = \frac{1}{2} \text{ 时坐标 } P = (0, -2, \frac{3\pi}{4})$$

$$\text{找法平面3条. 设 } Q(x, y, z) \quad PQ \perp T \quad \therefore PQ \cdot T = 0$$

$$PQ = (x, y+2, z-\frac{3\pi}{4}) \quad T(-\pi, 0, 3)$$

$$\text{法平面3条: } -\pi x + 3(z - \frac{3\pi}{4}) = 0$$

$$\text{切线3条: } \frac{x-x_p}{x'} = \frac{y-y_p}{y'} = \frac{z-z_p}{z'}$$

$$= \frac{x}{-\pi} = \frac{y+2}{0} = \frac{z-\frac{3\pi}{4}}{3}$$

6. Find the point P on the plane $3x + y + z - 2 = 0$ closest to the origin $O(0, 0, 0)$ by using the

Lagrange multiplier method. Answer for the auxiliary function _____

Answer for point P _____

拉格朗日乘数法.

$$\left\{ \begin{array}{l} \text{设 } P(x, y, z) \\ OP = |d| \Rightarrow d^2 = x^2 + y^2 + z^2 \end{array} \right.$$

若为最近点 (a, b, c) 点

$$\downarrow \text{例 } d^2 = (x-a)^2 + (y-b)^2 + (z-c)^2$$

$$\text{辅助函数 } F(x, y, z, \lambda) = x^2 + y^2 + z^2 + \lambda(3x + y + z - 2)$$

$$\frac{\partial F}{\partial x} = 2x + 3\lambda = 0$$

$$\frac{\partial F}{\partial y} = 2y + \lambda = 0$$

$$\frac{\partial F}{\partial z} = 2z + \lambda = 0$$

若 z 有表达式
e.g. $z = x^2 + y^2$
 $\lambda + \lambda_2(x^2 + y^2)$

$$\frac{\partial F}{\partial \lambda} = 3x + y + z - 2 = 0$$

$$\lambda - 2y - 2z = -\frac{2x}{3}$$

$$= 3\left(-\frac{3\lambda}{2}\right) - \frac{\lambda}{2} - \frac{\lambda}{2} - 2 = 0$$

$$\lambda = \frac{4}{11}$$

$$\therefore x = \frac{6}{11} \quad y = \frac{2}{11} \quad z = \frac{2}{11}$$

$$\therefore P\left(\frac{6}{11}, \frac{2}{11}, \frac{2}{11}\right)$$

7. The plane $x + y + 2z = 2$ intersects the paraboloid $z = x^2 + y^2$ in an ellipse. Find the points on the ellipse that are nearest to and farthest from the origin.

椭圆

Answer for the nearest point: $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$

Answer for the farthest point: $(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$

$$\text{距离}^2 = L^2 = x^2 + y^2 + z^2$$

$$\text{约束条件} \begin{cases} x + y + 2z - 2 = 0 \\ z - x^2 - y^2 = 0 \end{cases}$$

$$\Rightarrow L(x, y, z, \lambda, \mu)$$

$$L = x^2 + y^2 + z^2 + \lambda(x + y + 2z - 2) + \mu(z - x^2 - y^2)$$

$$\begin{cases} \frac{\partial L}{\partial x} = 2x + \lambda - 2\mu x = 0 \\ \frac{\partial L}{\partial y} = 2y + \lambda - 2\mu y = 0 \\ \frac{\partial L}{\partial z} = 2z + 2\lambda - \mu = 0 \\ \frac{\partial L}{\partial \lambda} = x + y + 2z - 2 = 0 \\ \frac{\partial L}{\partial \mu} = z - x^2 - y^2 = 0 \end{cases}$$

$$\Rightarrow x = y$$

代入的得到 x, y, z 关系

$$\Rightarrow z = x^2 + y^2 = 2x^2$$

$$\text{代入 plane: } \begin{aligned} x + x + 2 \cdot 2x^2 &= 2 \\ 2x + 4x^2 &= 2 \\ 2x^2 + x - 1 &= 0 \end{aligned}$$

$$x = \frac{-1 \pm 3}{4}$$

$$x_1 = \frac{1}{4}, x_2 = -\frac{1}{2}$$

$$P_1(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}) \quad P_2(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$$

$$d_1 = \sqrt{\frac{3}{12}}$$

$$d_2 = \sqrt{\frac{3}{4}}$$

$$d_1 < d_2$$

8. Find the points on the ellipsoid $x^2 + 2y^2 + 3z^2 = 1$ where the tangent plane is parallel to the plane $3x - y + 3z - 5 = 0$. Your answer: $\pm(\frac{3\sqrt{2}}{5}, \frac{\sqrt{2}}{10}, \frac{\sqrt{2}}{5})$ 2个点. 入为 $\pm \frac{\sqrt{2}}{5}$.

$$f(x, y, z) = x^2 + 2y^2 + 3z^2$$

切平面

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$= (2x, 4y, 6z)$$

给定平面的法向量 $\vec{n} = (3, -1, 3)$

$$\text{找点 } (x_0, y_0, z_0) \quad \nabla f(x_0, y_0, z_0) = k \cdot \vec{n}$$

$$2x_0 = 3k \quad x_0 = \frac{3}{2}k = \pm \frac{3\sqrt{2}}{5}$$

$$4y_0 = -1k \quad y_0 = -\frac{1}{4}k = \pm \frac{\sqrt{2}}{10}$$

$$6z_0 = 3k \quad z_0 = \frac{1}{2}k = \pm \frac{\sqrt{2}}{5}$$

$$\text{代入 } x_0^2 + 2y_0^2 + 3z_0^2 = 1$$

$$\frac{9}{4}k^2 + \frac{1}{8}k^2 + \frac{3}{4}k^2 = 1$$

$$\frac{25}{8}k^2 = 1 \quad k^2 = \frac{8}{25}$$

$$k = \pm \frac{2\sqrt{2}}{5}$$

9. Calculate the iterated integrals

$$\int_0^1 \left(\int_x^1 \sin(y^2) dy \right) dx =$$

先画图写出双重积分, 再交换积分顺序

交换积分次序

$$\int_0^1 dx \int_x^1 \sin(y^2) dy$$

$$= \int_0^1 dy \int_0^y \sin(y^2) dx$$

$$= \int_0^1 y \sin(y^2) dy$$

$$= -\frac{1}{2} \cos(y^2) \Big|_0^1$$

$$= \frac{1 - \cos 1}{2}$$

$$\begin{aligned} x: 0 \text{ to } 1 \\ y: x \text{ to } 1 \end{aligned} \Rightarrow \begin{aligned} y: 0 \text{ to } 1 \\ x: y \text{ to } 1 \end{aligned}$$

10. Suppose that $f(x, y) = \sqrt{1 - x^2 - y^2} - \frac{2}{\pi} \iint_D f(x, y) dx dy$, and $D = \{(x, y) | x^2 + y^2 \leq 1\}$. Evaluate

the integral $\iint_D f(x, y) dx dy = \frac{2\pi}{9}$

设 $M = \iint_D f(x, y) dx dy$

$$\iint_D f(x, y) dx dy = \iint_D \sqrt{1 - x^2 - y^2} dx dy - \frac{2}{\pi} \iint_D M dx dy$$

$$\therefore M = \iint_D \sqrt{1 - x^2 - y^2} dx dy - \frac{2}{\pi} M \pi$$

$$S = \pi r^2 = \pi \quad (x^2 + y^2 < 1) \text{ 圆 } R=1$$

$$\iint_D \sqrt{1 - x^2 - y^2} dx dy$$

$$= \int_0^{2\pi} d\theta \int_0^1 r \sqrt{1 - r^2} dr \quad r^2 = x^2 + y^2$$

$$= \int_0^{2\pi} d\theta \int_0^1 \left(-\frac{1}{2}\right) \sqrt{1 - r^2} d(1 - r^2)$$

$$= \int_0^{2\pi} d\theta \left[-\frac{2}{3} \left(-\frac{1}{2}\right) \cdot (1 - r^2)^{\frac{3}{2}} \right]_0^1$$

$$= -\frac{1}{3} \cdot 2\pi \cdot (0 - 1)$$

$$= \frac{2\pi}{3}$$

$$\therefore M = \frac{2\pi}{3}$$

$$M = \frac{2\pi}{9}$$

$$M = \frac{2\pi}{3} - \frac{2}{\pi} M \pi = \frac{2\pi}{3} - 2M$$

11. The plane region D is such that the double integral $\iint_D \sqrt{1-x^2-y^2} dx dy$ achieves the maximum.

Then the plane region D is 圆域

and the maximum integral equals 保证

唯一的函数 $f(x,y) > 0$

由上题 $0 \leq x^2 + y^2 \leq 1$

$$D = \{(x, y) \mid x^2 + y^2 \leq 1\}$$

$$\underline{\text{Max} = \frac{2\pi}{3}}$$

12. Find the directions in which the function $f(x, y) = x^2 + \sin(xy)$

(a) decreases most rapidly at the point $P(1, 0)$.

Answer:

$$\begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

(b) has directional derivative equal to 1 at the point $P(1, 0)$.

Answer:

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \end{pmatrix}$$

3向导数 =
下降最快.

$$\cos \theta = \frac{4}{5}$$

$$\begin{matrix} 5 \\ 4 \end{matrix} \begin{matrix} 3 \\ 4 \end{matrix}$$

a. $f(x, y) = x^2 + \sin(xy)$

$$\frac{\partial f}{\partial x} = 2x + y \cos(xy)$$

$$\frac{\partial f}{\partial y} = x \cos(xy)$$

at point $P(1, 0)$ $\frac{\partial f}{\partial x} = 2$ $\frac{\partial f}{\partial y} = 1$

$$\therefore \nabla f(1, 0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

下降最快方向为梯度的负方向.

$$\therefore -\nabla f(1, 0) = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

b. 3向导数是梯度和3向向量的点积

$$\nabla f(1, 0) \cdot \vec{u} = 1$$

设 $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$

$$\therefore 2u_1 + u_2 = 1$$

设 $u_1 = \cos \theta$ $u_2 = \sin \theta - \frac{2}{3}$

$$2\cos \theta + \sin \theta = 1$$

$$\sin \theta = 1 - 2\cos \theta$$

① $\cos \theta = 0$ $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
② $\cos \theta = \frac{4}{5}$ $\theta = \arccos \frac{4}{5}$

$$(1 - 2\cos \theta)^2 + \cos^2 \theta = 1$$

$$1 + 4\cos^2 \theta - 4\cos \theta + \cos^2 \theta = 1$$

$$5\cos^2 \theta = 4\cos \theta$$

找其余代 $\sin^2 \theta + \cos^2 \theta = 1$

13. Use the Green's theorem to evaluate the line integral

$$\oint_C (5y - e^{\cos x}) dx + (6x + \sqrt{y^2 + 1}) dy$$

where C is the circle $x^2 + y^2 = 4$. Answer for the line integral is 4π

格林定理

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\oint_C \underbrace{(5y - e^{\cos x})}_{P} dx + \underbrace{(6x + \sqrt{y^2 + 1})}_{Q} dy$$

$$\frac{\partial P}{\partial y} = 5 \quad \frac{\partial Q}{\partial x} = 6$$

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = \iint_D 6 - 5 dA = \iint_D 1 dA$$

$$S = \pi r^2 = 4\pi$$

$$\therefore \iint_D 1 dA = S = 4\pi$$

14. Use symmetry to compute the following double integral

$$\iint_{D: x^2+y^2 \leq 1} (3x^2 \tan x + \sin(x^2 y) + 4y^5 + 1) dx dy.$$

Answer for the double integral is

π

奇
奇
奇
奇

求面积

$$D: (x^2 + y^2 \leq 1) \quad \text{圆} \quad r=1$$

$$\text{设: } -\sqrt{1-y^2} \leq x \leq \sqrt{1-y^2} \quad -1 \leq y \leq 1$$

$$\int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} (3x^2 \tan x + \sin(x^2 y) + 4y^5 + 1) dx dy$$

$$\int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 1 dx dy$$

$$= \iint_D 1 d\sigma$$

$$= \pi r^2 = \pi$$

15. The vector field $\mathbf{F}(x, y, z) = y \cos(xy)\mathbf{i} + x \cos(xy)\mathbf{j} - \sin z\mathbf{k}$ is a conservative vector field satisfying $\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}$. Find a scalar function $f(x, y, z)$ such that $\mathbf{F} = \nabla f$.

∇f 为 f 的梯度

Answer for the potential function: $f(x, y, z) = \sin xy + \cos z + C$

向量场

$$\frac{\partial f}{\partial x} = y \cos(xy)$$

x 部分

$$h(x) = \int y \cos(xy) = \sin(xy) + g(y, z)$$

y 部分

$$\frac{\partial f}{\partial y} = x \cos(xy)$$

$$h(y) = \int x \cos(xy) = \sin(xy) + h(z)$$

z 部分

$$h(z) = 1 - \sin z = \cos z$$

$$\therefore f(x, y, z) = \sin(xy) + \cos z + C$$

e.g. $F(x, y, z) = y^2 z^3 \vec{i} + 2xy z^3 \vec{j} + 3xy^2 z^2 \vec{k}$

① curl $\vec{F} = 0$ ② $f(x, y, z) \rightarrow \vec{F} = \nabla f$

$$f_x(x, y, z) = y^2 z^3$$

$$\frac{\partial f_x}{\partial y} = 2yz^3$$

$$\frac{\partial f_x}{\partial z} = 3y^2 z^2$$

$$f_y(x, y, z) = 2xyz^3$$

$$\frac{\partial f_y}{\partial x} = 2yz^3$$

$$\frac{\partial f_y}{\partial z} = 6xyz^2$$

$$f_z(x, y, z) = 3xy^2 z^2$$

$$\frac{\partial f_z}{\partial x} = 3y^2 z^2$$

$$\frac{\partial f_z}{\partial y} = 6xy z^2$$

$$\frac{\partial f}{\partial x} = y^2 z^3$$

$$\int \frac{\partial f}{\partial x} = xy^2 z^3 \quad \text{X } \frac{3}{y} \text{ 部分}$$

$$\therefore f = xy^2 z^3 + g(y, z)$$

$$\text{Curl } \vec{F} = \left(\frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z} \right) \vec{i} + \left(\frac{\partial f_z}{\partial x} - \frac{\partial f_x}{\partial z} \right) \vec{j}$$

$$+ \left(\frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y} \right) \vec{k}$$

$$= 0$$

$$\int \frac{\partial f}{\partial y} = xy^2 z^3 \quad \text{y 部分}$$

$$g(y, z) = h(z)$$

$$\int \frac{\partial f}{\partial z} = xy^2 z^3 \quad \text{z 部分}$$

$$h'(z) = 0 \quad h(z) = C$$

$$\therefore f(x, y, z) = xy^2 z^3 + C$$

16. (10 marks) First express the following triple integral as iterated integrals in spherical coordinate system, and then evaluate the triple integral.

$$I = \iiint_{\Omega} (x^2 + y^2 + z^2) dx dy dz,$$

where $\Omega = \{(x, y, z) | x^2 + y^2 + z^2 \leq 1, x \geq 0, y \geq 0, \text{ and } z \geq 0\}$.

位于第一象限单位球体

$$\text{in } \Sigma x = r \sin \varphi \cos \theta$$

$$y = r \sin \varphi \sin \theta$$

$$z = r \cos \varphi$$

由于在第一象限

$$0 \leq \varphi \leq \frac{\pi}{2}, \quad 0 \leq \theta \leq 2\pi$$

$$J = r^2 \sin \varphi$$

$$\therefore I = \iiint_{\Omega} r^2 \cdot r^2 \sin \varphi dr d\varphi d\theta$$

$$= \int_0^1 dr \int_0^{\frac{\pi}{2}} d\varphi \int_0^{\frac{\pi}{2}} r^4 \sin \varphi d\varphi$$

$$= \frac{\pi}{2} \int_0^1 dr (r^4 \cos \varphi) \Big|_{\varphi=0}^{\varphi=\frac{\pi}{2}}$$

$$= \frac{\pi}{2} \int_0^1 -r^4 (0 - 1) dr$$

$$= \frac{\pi}{2} \int_0^1 r^4 dr$$

$$= \frac{\pi}{2} \cdot \frac{1}{5} r^5 \Big|_0^1$$

$$= \frac{\pi}{10}$$

17. (10 marks) Find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 - z = 0$. Find in $(0, 0, \frac{1}{2})$

球坐標系 - 球面坐標系

$$\rho^2 = \rho \cos \varphi \quad \rho = \cos \varphi$$

$$z = \sqrt{x^2 + y^2}$$

$$\begin{cases} x = \rho \sin \varphi \cos \theta \\ y = \rho \sin \varphi \sin \theta \\ z = \rho \cos \varphi \end{cases}$$

$$\rho \cos \varphi = \rho \sin \varphi$$

$$\cos \varphi = \sin \varphi$$

$$\varphi = \frac{\pi}{4}$$

$$\therefore \Omega = \{(\rho, \varphi, \theta) \mid 0 \leq \varphi \leq \frac{\pi}{4}, 0 \leq \theta \leq 2\pi, 0 \leq \rho \leq \cos \varphi\}$$

$$V = \iiint_{\Omega} \rho \, dV = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^{\cos \varphi} \rho^2 \sin \varphi \, d\rho$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \cdot \frac{1}{3} \cos^3 \varphi \sin \varphi$$

$$= \frac{2\pi}{3} \left[-\frac{1}{4} \cos^4 \varphi \right]_0^{\frac{\pi}{4}}$$

$$= -\frac{\pi}{6} \left(\left(\frac{\sqrt{2}}{2} \right)^4 - 1 \right)$$

$$= -\frac{\pi}{6} \times -\frac{3}{4}$$

$$= \frac{\pi}{8}$$

$$\frac{2}{4} \times \frac{3}{4} = \frac{3}{4}$$

18. (10 marks) Express the following triple integral as iterated integrals in cylindrical coordinates, and then find the result $I = \iiint_{\Omega} z \, dv$, where Ω is the region bounded by the paraboloid $z = x^2 + y^2$ and the hemisphere $z = \sqrt{6 - x^2 - y^2}$.

交线: $z = x^2 + y^2 = \sqrt{6 - x^2 - y^2}$

平方: $(x^2 + y^2)^2 = 6 - x^2 - y^2$

$(x^2 + y^2)^2 + (x^2 + y^2) - 6 = 0$

let $t = x^2 + y^2 \therefore t^2 + t - 6 = 0$

$(t+3)(t-2) = 0 \quad \begin{matrix} t = -3 \\ t = 2 \end{matrix}$

$t_1 = -3$ (舍) $t_2 = 2 = x^2 + y^2 \therefore r = \sqrt{2}$ (圆)

$\begin{cases} z^2 = x^2 + y^2 \\ x = r \cos \theta \\ y = r \sin \theta \end{cases}$

原式 $\int_0^{\pi} d\theta \int_0^{\sqrt{2}} r \, dr \int_{x^2+y^2}^{\sqrt{6-x^2-y^2}} z \, dz$

$= \int_0^{\pi} d\theta \int_0^{\sqrt{2}} r \, dr \cdot \frac{1}{2} (6 - x^2 - y^2 - (x^2 + y^2)^2)$

$= \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} r \cdot \frac{1}{2} (6 - r^2 - r^4) \, dr$

$= \frac{1}{2} \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} (6r - r^3 - r^5) \, dr$

$= \frac{1}{2} \cdot 2\pi \left(3r^2 - \frac{1}{4}r^4 - \frac{1}{6}r^6 \right) \Big|_0^{\sqrt{2}}$

$= \pi \left(6 - 1 - \frac{4}{3} \right)$

$= \frac{11}{3} \pi$

19. (10 marks) Let $F(x, y) = (-y\mathbf{i} + x\mathbf{j})/(x^2 + y^2)$. Show that $\oint_C F \cdot d\mathbf{r} = 2\pi$ for every positively oriented simple closed path that encloses the origin.

格林定理 不适用

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

$$F(x, y) = \frac{-y}{x^2 + y^2} \mathbf{i} + \frac{x}{x^2 + y^2} \mathbf{j}$$

Assume $\frac{1}{2} \leq x^2 + y^2 \leq 1$

$$x = r \cos \theta \quad y = r \sin \theta \quad 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi$$

$$dx = -\sin \theta d\theta \quad dy = \cos \theta d\theta$$

$$\begin{aligned} \therefore F \cdot d\mathbf{r} &= \left(\frac{-y\mathbf{i} + x\mathbf{j}}{x^2 + y^2} \right) \cdot (-\sin \theta d\theta \mathbf{i} + \cos \theta d\theta \mathbf{j}) \\ &= (-\sin \theta \mathbf{i} + \cos \theta \mathbf{j}) \cdot (-\sin \theta d\theta \mathbf{i} + \cos \theta d\theta \mathbf{j}) \\ &= \sin^2 \theta d\theta + \cos^2 \theta d\theta \\ &= d\theta \end{aligned}$$

$$\therefore \oint_C F \cdot d\mathbf{r} = \int_0^{2\pi} d\theta = 2\pi$$

2.
Proof: $F(x, y) = \frac{-y}{x^2 + y^2} \mathbf{i} + \frac{x}{x^2 + y^2} \mathbf{j}$
 $\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$
 let $P(x, y) = \frac{-y}{x^2 + y^2}$, $Q(x, y) = \frac{x}{x^2 + y^2}$
 $\int_C P dx + Q dy = \int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D \left[\frac{x^2 + y^2 - x^2}{(x^2 + y^2)^2} - \frac{y^2 - y^2}{(x^2 + y^2)^2} \right] dA = 0$
 $\therefore \int_C P dx + Q dy = \int_C P dx + Q dy$
 $\Rightarrow \int_C F \cdot d\mathbf{r} = \int_C F \cdot d\mathbf{r}$
 let $\vec{r}(t) = a \cos t \mathbf{i} + a \sin t \mathbf{j}$, $0 \leq t \leq 2\pi$
 $\vec{r}'(t) = -a \sin t \mathbf{i} + a \cos t \mathbf{j}$
 $\vec{F}(\vec{r}(t)) = -\frac{\sin t}{a} \mathbf{i} + \frac{\cos t}{a} \mathbf{j}$
 $\therefore \int_C \vec{F} \cdot d\mathbf{r} = \int_0^{2\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$
 $= \int_0^{2\pi} 1 \cdot dt$
 $= 2\pi$

20. (Bonus problem) Use the divergence theorem to evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = \langle xy, e^{xz^2} + y^2, xy \cos(xy) \rangle$ and S is the surface of the region E bounded by the parabolic cylinder $z = 1 - x^2$ and the planes $z = 0$, $y = 0$ and $y + z = 2$.

附加题: $\text{div } \mathbf{F} = \frac{\partial(xy)}{\partial x} + \frac{\partial(e^{xz^2} + y^2)}{\partial y} + \frac{\partial(xy \cos(xy))}{\partial z}$

$$= y$$

求面积分

$$\left\{ \begin{array}{l} \text{在 } z = 1 - x^2 \text{ 上} \\ z = 0 \\ y = 0 \\ y + z = 2 \end{array} \right.$$

分别求面积分

相加